02LQD Specification and Simulation of Digital Systems

2021/02/05

1. Combinational circuit design (2 points)

A museum has three rooms, each with a motion sensor (m0, m1, and m2) that outputs 1 when motion is detected. At night, the only person in the museum is one security guard who walks from room to room. Create a circuit that sounds an alarm (by setting an output A to 1) if motion is ever detected in more than one room at a time (i.e., in two or three rooms), meaning there must be one or more intruders in the museum.

Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (16 points)

A Mealy FSM has a 1-bit input x and a 1-bit output z. Output z is 1 if and only if the most recent input, combined with the preceding three inputs, was not a valid BCD encoding of a decimal digit; otherwise, z = 0. Assume the BCD digits are received least significant bit first. Assume that in the initial state all previous inputs were 0. Reset is synchronous.

3. Draw the state diagram (**4 points**)
4. Design the FSM resorting to the VHDL behavioural description style with 2 processes (**4 points**).
5. Design the FSM resorting to the VHDL behavioural description style with 1 process (**4 points**).
6. Design the FSM resorting in Verilog with 2 always blocks (**4 points**).

3. HLSM design (12 points)

Jacobsthal numbers are recursively defined as follows:

$J_n = J_{n-1} + 2J_{n-2}$ $\qquad$ $n \geq 2$

$J_0 = 0$

$J_1 = 1$

The first 11 Jacobsthal numbers are thus: 0, 1, 1, 3, 5, 11, 21, 43, 85, 171, 341.

Design a HLSM to **iteratively** compute Jacobsthal numbers of order n, where n is a 4-bit input. Computation starts when a start signal is asserted for one clock cycle. When computation is over the ready signal is asserted for one clock cycle. Reset is asynchronous. The output is J. Define the appropriate size for J. The output J makes sense only when ready is 1.

Draw the HLSM state diagram (**6 points**) and write the VHDL code (**6 points**) to capture the HLSM behavior.

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
  - running VHDL code and testbench
  - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

  no later than Monday February 8 11.59 pm
- format of .zip archive name surname_ID (example Rossi_244123)
- remember: no upload ⇒ no correction ⇒ no oral exam
- once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.